

FORMELSAMMLUNG TECHNISCHE MECHANIK II

I. Spannung, Verzerrung & Elastizitätsgesetze

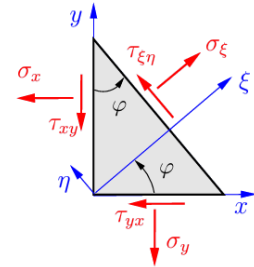
I.1 Spannungszustände

(i) Transformationsgleichungen

$$\sigma_{\xi} = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y) \cos 2\varphi + \tau_{xy} \sin 2\varphi$$

$$\sigma_{\eta} = \frac{1}{2}(\sigma_x + \sigma_y) - \frac{1}{2}(\sigma_x - \sigma_y) \cos 2\varphi - \tau_{xy} \sin 2\varphi$$

$$\tau_{\xi\eta} = -\frac{1}{2}(\sigma_x - \sigma_y) \sin 2\varphi + \tau_{xy} \cos 2\varphi$$

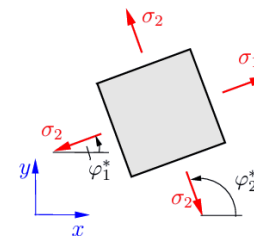


(ii) Hauptspannungen

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tan 2\varphi^* = \frac{\tau_{xy}}{\sigma_x - \sigma_y}; \quad \varphi_2^* = \varphi_1^* \pm \frac{\pi}{2}$$

$$\sigma_M = \frac{1}{2}(\sigma_x + \sigma_y) = \frac{1}{2}(\sigma_1 + \sigma_2)$$

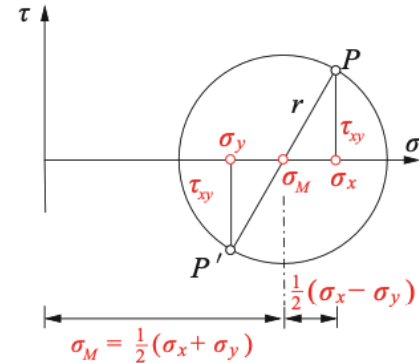


(iii) Maximale Schubspannung

$$\tau_{max} = \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}; \quad \tan 2\varphi^{**} = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}};$$

Extremwert bei $\varphi = \varphi^{**}$

$$\tau_{max} = \pm \frac{1}{2}(\sigma_1 - \sigma_2)$$



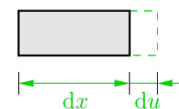
I.2 Verzerrungszustände

(i) Einachsiger Verzerrungszustand

$$\varepsilon(x) = \frac{du}{dx} = \frac{\Delta l}{l}$$

(ii) Zweiachsiger Verzerrungszustand

$$\varepsilon_x = \frac{du}{dx}; \quad \varepsilon_y = \frac{dv}{dy}; \quad 2\varepsilon_{xy} = \gamma_{xy} = \frac{du}{dy} + \frac{dv}{dx} = \gamma_{yx}$$

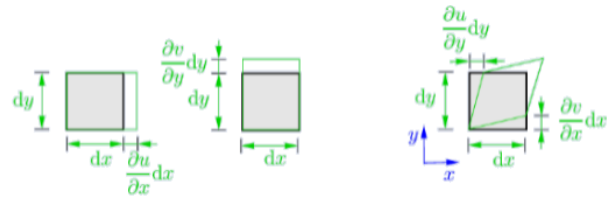


(iii) Transformationsgleichungen

$$\varepsilon_{\xi} = \frac{1}{2}(\varepsilon_x + \varepsilon_y) + \frac{1}{2}(\varepsilon_x - \varepsilon_y) \cos 2\varphi + \frac{1}{2}\gamma_{xy} \sin 2\varphi$$

$$\varepsilon_{\eta} = \frac{1}{2}(\varepsilon_x + \varepsilon_y) - \frac{1}{2}(\varepsilon_x - \varepsilon_y) \cos 2\varphi - \frac{1}{2}\gamma_{xy} \sin 2\varphi$$

$$\frac{1}{2}\gamma_{\xi\eta} = -\frac{1}{2}(\varepsilon_x - \varepsilon_y) \sin 2\varphi + \frac{1}{2}\gamma_{xy} \cos 2\varphi$$



(iv) Hauptdehnungen

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{1}{2}\gamma_{xy}\right)^2} ; \quad \tan 2\varphi^* = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} ; \quad \varphi_2^* = \varphi_1^* \pm \frac{\pi}{2}$$

I.3 Elastizitätsgesetze

(i) Kennwerte

E : Elastizitätsmodul

ν : Querkontraktionszeit

G : Schubmodul

$$\text{sodass: } G = \frac{E}{2(1+\nu)}$$

(ii) Einachsiger Spannungszustand

$$\varepsilon = \frac{\sigma}{E} + \alpha_T \Delta T$$

(iii) Dreiachsiger Spannungszustand

$$\varepsilon_x = \frac{1}{E}[\sigma_x - \nu(\sigma_y + \sigma_z)] + \alpha_T \Delta T; \quad \gamma_{xy} = \frac{1}{G}\tau_{xy}$$

$$\varepsilon_y = \frac{1}{E}[\sigma_y - \nu(\sigma_z + \sigma_x)] + \alpha_T \Delta T; \quad \gamma_{yz} = \frac{1}{G}\tau_{yz}$$

$$\varepsilon_z = \frac{1}{E}[\sigma_z - \nu(\sigma_x + \sigma_y)] + \alpha_T \Delta T; \quad \gamma_{zx} = \frac{1}{G}\tau_{zx}$$

II. Zug & Druck in Stäben

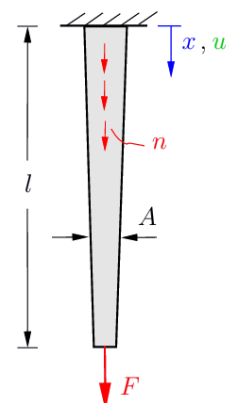
$$\sigma(x) = \frac{N(x)}{A(x)} = \frac{F}{A}$$

Für $EA = \text{konst.}$ und $\Delta T = \text{konst.}$ gelten

(i) DGL der Verschiebung

$$EAu' = N(x)$$

$$EAu'' = -n$$



(ii) Längenänderung

$$\Delta l = \int_0^l \left(\frac{N}{EA} + \alpha_T \Delta T \right) dx$$

III. Balken

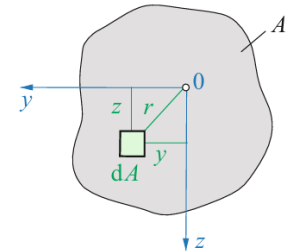
III.1 Geometrische Eigenschaften des Querschnittes

(i) Flächenträgheitsmomente

$$I_y = \int z^2 dA; \quad I_z = \int y^2 dA$$

$$I_{yz} = I_{zy} = - \int yz dA$$

$$I_p = \int r^2 dA = \int (z^2 + y^2) dA = I_y + I_z$$

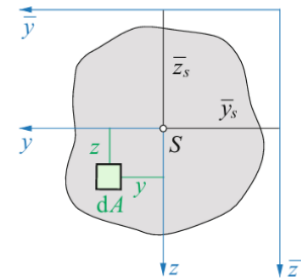


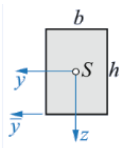
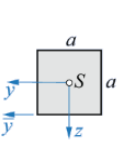
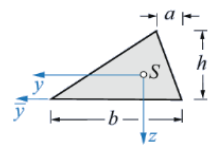
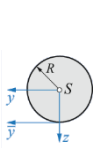
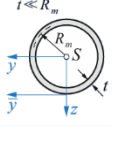
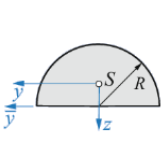
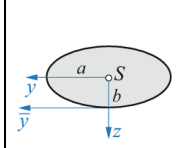
(ii) Satz von Steiner

$$I_{\bar{y}} = I_y + \bar{z}_S^2 A; \quad I_{\bar{z}} = I_z + \bar{y}_S^2 A$$

$$I_{\bar{y}\bar{z}} = I_{yz} - \bar{y}_S \bar{z}_S A$$

$$I_{\bar{p}} = I_p + \bar{r}_S^2 A$$



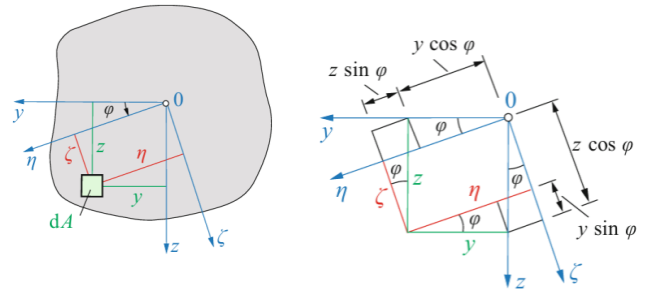
	Rechteck	Quadrat	Dreieck	Kreis	Dünner Kreis- ring	Halbkreis	Ellipse
							
I_y	$\frac{bh^3}{12}$	$\frac{a^4}{12}$	$\frac{bh^3}{36}$	$\frac{\pi R^4}{4}$	$\pi R_m^3 t$	$\frac{R^4}{72\pi} (9\pi^2 - 64)$	$\frac{\pi}{4} ab^3$
I_z	$\frac{hb^3}{12}$	$\frac{a^4}{12}$	$\frac{bh}{36} (b^2 - ba + a^2)$	$\frac{\pi R^4}{4}$	$\pi R_m^3 t$	$\frac{\pi R^4}{8}$	$\frac{\pi}{4} ba^3$
I_{yz}	0	0	$-\frac{bh^2}{72} (b - 2a)$	0	0	0	0
I_p	$\frac{bh}{12} (h^2 + b^2)$	$\frac{a^4}{6}$	$\frac{bh}{36} (h^2 + b^2 - ba + a^2)$	$\frac{\pi R^4}{2}$	$2\pi R_m^3 t$	$\frac{R^4}{36\pi} (9\pi^2 - 32)$	$\frac{\pi ab}{4} (a^2 + b^2)$
$I_{\bar{y}}$	$\frac{bh^3}{3}$	$\frac{a^4}{3}$	$\frac{bh^3}{12}$	$\frac{5\pi}{4} R^4$	$3\pi R_m^3 t$	$\frac{\pi R^4}{8}$	$\frac{5\pi}{4} ab^3$

(iii) Transformationsgleichungen

$$I_{\eta} = \frac{1}{2}(I_y + I_z) + \frac{1}{2}(I_y - I_z) \cos 2\varphi + I_{yz} \sin 2\varphi$$

$$I_{\zeta} = \frac{1}{2}(I_y + I_z) - \frac{1}{2}(I_y - I_z) \cos 2\varphi - I_{yz} \sin 2\varphi$$

$$I_{\eta\zeta} = -\frac{1}{2}(I_y - I_z) \sin 2\varphi + I_{yz} \cos 2\varphi$$

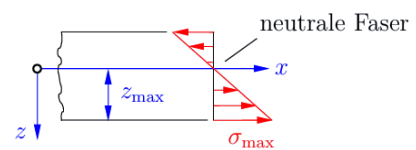


III.2 Balkenbiegung

(i) Gerade Biegung

$$\sigma = \frac{M}{I} z$$

$$\sigma_{max} = \frac{M}{W}, \quad \text{mit } W = \frac{I}{|z_{max}|}$$



(ii) Schiefe Biegung + Normalkraft

$$\sigma = \frac{N}{A} + \frac{1}{\Delta} [(M_y I_z - M_z I_{yz}) z - (M_z I_y - M_y I_{yz}) y]$$

$$\text{mit } \Delta = I_y I_z - I_{yz}^2$$

(iii) DGL der Biegelinie

$$-M(x) = EI w''$$

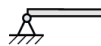
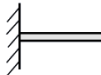
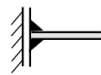
$$-Q(x) = (EI w'')$$

$$q(x) = (EI w'')''$$

Für $EI = \text{konst.}$ gilt

$$q(x) = EI w^{IV}$$

Statische Randbedingungen

Lager	w	w'	M	Q
	0	$\neq 0$	0	$\neq 0$
	0	0	$\neq 0$	$\neq 0$
freier Rand	$\neq 0$	$\neq 0$	0	0
	$\neq 0$	0	$\neq 0$	0

(iv) FÖPPL-Symbol

$$\langle x - a \rangle^n = \begin{cases} 0 & \text{für } x < a, \\ (x - a)^n & \text{für } x > a \end{cases}$$

$q = q_0 \langle x - a \rangle^0$	$q = \frac{q_b}{b} \langle x - a \rangle^1$	$Q = -F \langle x - a \rangle^0$	$M = -M_0 \langle x - a \rangle^0$	$q = q_0 - q_0 \langle x - a \rangle^0$

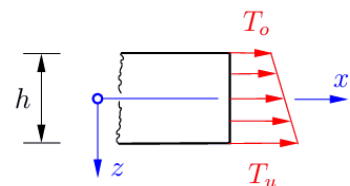
(v) Superposition der Biegelinie

Nr.	Lastfall	Elw'_A	Elw'_B	$Elw(x)$	Elw_{max}
1		$\frac{Fl^2}{6}(\beta - \beta^3)$	$-\frac{Fl^2}{6}(\alpha - \alpha^3)$	$\frac{Fl^3}{6}[\beta\xi(1 - \beta^2 - \xi^2) + \langle \xi - \alpha \rangle^3]$	$\frac{Fl^3}{48}$ für $\alpha = \beta = 1/2$
2		$\frac{q_0 l^3}{24}$	$-\frac{q_0 l^3}{24}$	$\frac{q_0 l^3}{24}(\xi - 2\xi^3 + \xi^4)$	$\frac{5}{384}q_0 l^4$
4		$\frac{7}{360}q_B l^3$	$-\frac{1}{45}q_B l^3$	$\frac{q_B l^4}{360}(7\xi - 10\xi^3 + 3\xi^5)$	
4		$\frac{M_0 l}{6}(3\beta^2 - 1)$	$\frac{M_0 l}{6}(3\alpha^2 - 1)$	$\frac{M_0 l^2}{6}[\xi(3\beta^2 - 1) + \xi^3 - 3\langle \xi - \alpha \rangle^2]$	$\frac{M_0 l^2}{27}\sqrt{3}$ für $a = 0$
5		0	$\frac{Fa^2}{2}$	$\frac{Fl^2}{6}[3\xi^2\alpha - \xi^3 + \langle \xi - \alpha \rangle^3]$	$\frac{Fl^3}{3}$ für $a = l$
6		0	$\frac{q_0 l^3}{6}$	$\frac{q_0 l^4}{24}(6\xi^2 - 4\xi^3 + \xi^4)$	$\frac{q_0 l^4}{8}$
7		0	$\frac{q_A l^3}{24}$	$\frac{q_A l^4}{120}(10\xi^2 - 10\xi^3 + 5\xi^4 - \xi^5)$	$\frac{q_A l^4}{30}$
8		0	$M_0 l^2$	$M_0 \frac{x^2}{2}$	$M_0 \frac{l^2}{2}$

Abkürzungen: $\xi = \frac{x}{l}$; $\alpha = \frac{a}{l}$; $\beta = \frac{b}{l}$; $El = \text{const.}$; $w' = \frac{dw}{dx}$

(vi) Temperaturbelastung

$$w'' = -\frac{M}{EI} - \alpha_T \frac{T_u - T_o}{h}$$



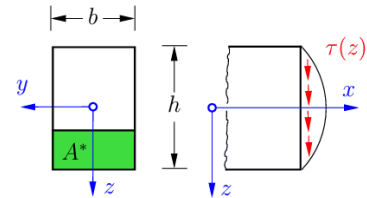
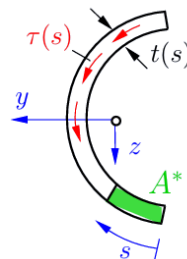
II.2 Schubspannung infolge der Querkraft

(i) Statische Momente

$$S_y = \int_A z \, dA; \quad S_z = \int_A y \, dA$$

(ii) Kompakte Querschnitte

$$\tau(z) = \frac{QS(z)}{Ib(z)}$$

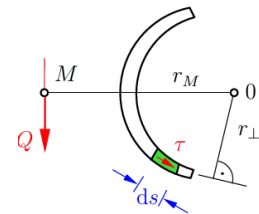


(iii) Dünnwandige, offene Profile

$$\tau(s) = \frac{QS(s)}{It(s)}$$

(iv) Schubmittelpunkt

$$r_M Q = \int \tau(s) r_\perp(s) t(s) \, ds$$



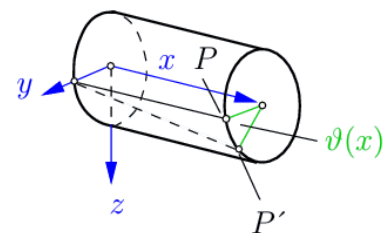
IV. Torsion

$$\tau_{max} = \frac{M_T}{W_T}; \quad \text{mit } W_T = \frac{I_T}{R}$$

(i) DGL der Verdrehung

$$M_T = GI_T \vartheta'$$

$$-m_T = (GI_T \vartheta')'$$



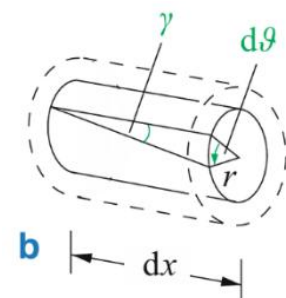
(ii) Verdrehung der Endquerschnitte

Für $GI_T = \text{konst.}$ und $M_T = \text{konst.}$ gilt

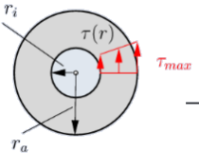
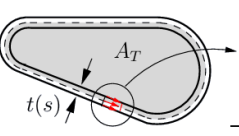
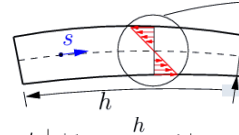
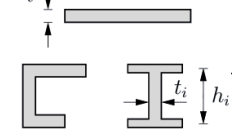
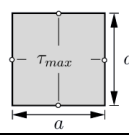
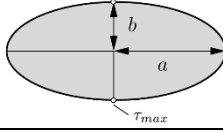
$$\Delta\vartheta = \frac{M_T l}{GI_T}$$

(iii) Kinematik der Torsion

$$r \, d\vartheta = \gamma \, dx$$



(iv) Grundformeln zur Torsion

Querschnitt	I_T	W_T
 $r_i = 0$ (Vollkreis)	$I_T = I_P = \frac{\pi}{2} (r_a^4 - r_i^4)$	$W_T = \frac{I_T}{r_a} = \frac{\pi}{2} \frac{r_a^4 - r_i^4}{r_a}$
 $a = \text{konst.}$ $t = \text{konst.}$	$I_T = \frac{4A_T^2}{\oint \frac{ds}{t(s)}}$	$W_T = 2A_T t_{\min}$
 $t = \text{konst.}$	$I_T = \frac{1}{3} \int_0^h t^3(s) ds$	$W_T = \frac{I_T}{t_{\max}}$
 $t_i = \text{konst.}$	$I_T = \frac{1}{3} \sum h_i t_i^3$	$W_T = \frac{I_T}{t_{\max}}$
 Quadrat	$I_T = 0,141a^4$	$W_T = 0,208a^3$
 Ellipse	$I_T = \pi \frac{a^3 b^3}{a^2 + b^2}$	$W_T = \frac{\pi}{2} ab^2$

(v) 1. Bredtsche Formel

$$\tau(s) = \frac{T}{t} = \frac{M_T}{2A_T t(s)}$$

(vi) 2. Bredtsche Formel

$$I_T = \frac{(2A_T)^2}{\oint \frac{ds}{t(s)}}$$

V. Der Arbeitsbegriff in der Elastostatik

V.1 Äußere Arbeit und gespeicherte Energie

(i) Arbeitssatz

$$W = \Pi$$

	Zug/Druck	Biegung	Torsion
Gleichgewicht	$N' = -n$	$M' - Q = 0$ $Q' = -q$	$M_T' = -m_T$
Kinematik	$\varepsilon = u'$	$\chi_B = -\psi'$ $\psi = -w'$	$\chi_T = -\vartheta'$
Elastizitätsgesetz	$N = EA\varepsilon$	$M = -EA\chi_B$	$M_T = GI_T\chi_T$
	$EAu'' = -n$	$EIw^{IV} = q$	$GI_T\vartheta'' = -m_T$

(ii) Äußere Arbeit

$$W = \int F df = \frac{1}{2} Ff \quad \text{mit } f: \text{ Verschiebung des Kraftangriffspunktes in Richtung der Kraft}$$

$$W = \int M d\varphi = \frac{1}{2} M\varphi \quad \text{mit } \varphi: \text{ Drehwinkel am Angriffspunkt in Richtung des Momentes}$$

(iii) Formänderungsenergie Π^* pro Längeneinheit

Zug/Druck	Biegung	Schub	Torsion
$\frac{1}{2} N\varepsilon$	$\frac{1}{2} M\psi'$	$\frac{1}{2} Q\tilde{\gamma}$	$\frac{1}{2} M_T\vartheta'$
$\frac{1}{2} EA\varepsilon^2$	$\frac{1}{2} EI\psi'^2$	$\frac{1}{2} GA_S\tilde{\gamma}^2$	$\frac{1}{2} GI_T\vartheta'^2$
$\frac{1}{2} \frac{N^2}{EA}$	$\frac{1}{2} \frac{M^2}{EI}$	$\frac{1}{2} \frac{Q^2}{GA_S}$	$\frac{1}{2} \frac{M_T^2}{GI_T}$

(iv) Gesamte Formänderungsenergie

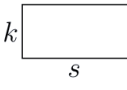
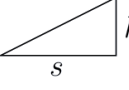
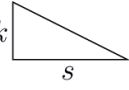
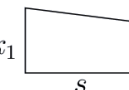
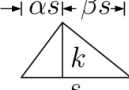
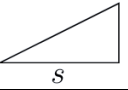
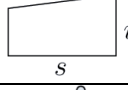
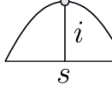
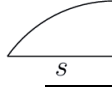
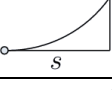
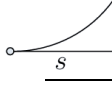
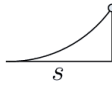
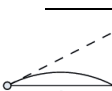
$$\Pi = \int_l \frac{N^2}{2EA} dx + \int_l \frac{M^2}{2EI} dx + \int_l \frac{Q^2}{2GA_S} dx + \int_l \frac{M_T^2}{2GI_T} dx$$

V.2 Prinzip der virtuellen Kräfte

(i) Verschiebung eines Punktes

$$f_i = \int_l \frac{N\bar{N}}{EA} dx + \int_l \frac{M\bar{M}}{EI} dx + \int_l \frac{Q\bar{Q}}{GA_S} dx + \int_l \frac{M_T\bar{M}_T}{GI_T} dx$$

(ii) Hilfstafel zur Ermittlung der Integrale $\int M_i M_k dx$

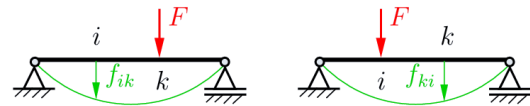
$M_i \backslash M_k$						quadratische Parabeln	
i	sik	$\frac{1}{2}sik$	$\frac{1}{2}sik$	$\frac{si}{2}(k_1 + k_2)$	$\frac{1}{2}sik$	$\frac{2}{3}sik$	$\frac{2}{3}sik$
	$\frac{1}{2}sik$	$\frac{1}{3}sik$	$\frac{1}{6}sik$	$\frac{si}{6}(k_1 + 2k_2)$	$\frac{1}{6}sik(1 + \alpha)$	$\frac{1}{3}sik$	$\frac{1}{4}sik$
	$\frac{s}{2}(i_1 + i_2)k$	$\frac{s}{6}(i_1 + 2i_2)k$	$\frac{s}{6}(2i_1 + i_2)k$	$\frac{s}{6}(2i_1k_1 + i_1k_2 + 2i_2k_2 + i_2k_1)$	$\frac{sk}{6}[(1 + \beta)i_1 + (1 + \alpha)i_2]$	$\frac{sk}{3}(i_1 + i_2)$	$\frac{sk}{12}(5i_1 + 3i_2)$
quadratische Parabel		$\frac{2}{3}sik$	$\frac{1}{3}sik$	$\frac{si}{3}(3k_1 + 5k_2)$	$\frac{1}{3}sik(1 + \alpha\beta)$	$\frac{8}{15}sik$	$\frac{7}{15}sik$
		$\frac{2}{3}sik$	$\frac{5}{12}sik$	$\frac{si}{12}(3k_1 + 5k_2)$	$\frac{sik}{12}(5 + \beta + \beta^2)$	$\frac{7}{15}sik$	$\frac{11}{30}sik$
		$\frac{1}{3}sik$	$\frac{1}{4}sik$	$\frac{si}{12}(k_1 + 3k_2)$	$\frac{sik}{12}(1 + \alpha + \alpha^2)$	$\frac{1}{5}sik$	$\frac{2}{15}sik$
kubische Parabel		$\frac{1}{4}sik$	$\frac{1}{5}sik$	$\frac{si}{20}(k_1 + 4k_2)$	$\frac{sik}{20}(1 + \alpha)(1 + \alpha^2)$	$\frac{2}{15}sik$	$\frac{1}{12}sik$
		$\frac{3}{8}sik$	$\frac{11}{40}sik$	$\frac{si}{40}(4k_1 + 11k_2)$	$\frac{sik}{10}\left(1 + \alpha + \alpha^2 - \frac{\alpha^3}{4}\right)$	$\frac{11}{15}sik$	$\frac{29}{120}sik$
		$\frac{1}{4}sik$	$\frac{2}{15}sik$	$\frac{si}{60}(7k_1 + 8k_2)$	$\frac{sik}{20}(1 + \alpha)\left(\frac{7}{3} - \alpha^2\right)$	$\frac{1}{5}sik$	$\frac{1}{6}sik$

Quadratische Parabeln: $\circ \triangleq$ Parabelscheitel,

Kubische Parabeln: $\circ \triangleq$ Nullstelle der Dreiecksbelastung

(iii) Vertauschungssatz von MAXWELL-BETTI

$$f_{ik} = f_{ki}$$

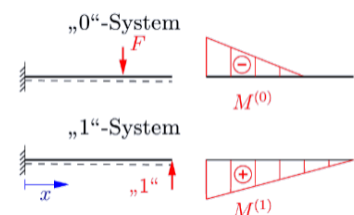
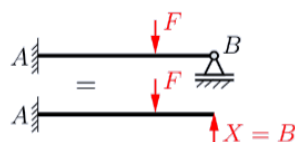


(iv) Anwendung bei statisch unbestimmten Problemen

$$X = -\frac{\alpha_{10}}{\alpha_{11}}$$

$$\text{mit } \alpha_{11} = \frac{\int M^{(1)} M^{(1)} dx}{EI}$$

$$\text{und } \alpha_{10} = \frac{\int M^{(0)} M^{(1)} dx}{EI}$$



VI. Knickung

VI.1 DGL des Eulerstabes (zweiter Ordnung)

(i) Knickgleichung

$$w'' + \lambda^2 w = 0 \quad \text{mit} \quad \lambda^2 = F/EI$$

(ii) Allg. Lösung der Knickgleichung

$$w = A \cos \lambda x + B \sin \lambda x$$

VI.2 DGL des Eulerstabes (vierter Ordnung)

(i) Knickgleichung

$$w^{IV} + \lambda^2 w'' = 0 \quad \text{mit} \quad \lambda^2 = F/EI$$

(ii) Allg. Lösung der Knickgleichung

$$w = A \cos \lambda x + B \sin \lambda x + C \lambda x + D$$

(iii) Kritische Last

$$F_{krit} = \pi^2 \frac{EI}{l_k^2}$$

